Chiyo (*Food is Medicine*) Lab

Research Indicates that Chiyo's Prenatal Nurture Program Improves Gut Microbiome Composition as well as Physical and Mental Health

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WHY WE DID THIS STUDY: OUR RESEARCH MISSION

At Chiyo, our research mission is to demonstrate and establish the vital role of evidence-based, holistic nutrition throughout the maternal health journey. As part of achieving this goal, we aim to understand how Chiyo's nutrition programs influence women's health across biological, psychological, and social dimensions across life stages. **This white paper presents preliminary results from a study that expands Chiyo's evidence base to the prenatal stage, demonstrating improvements in health across all of these dimensions.**

THE GUT MICROBIOME: A KEY CONDUIT OF GENERATIONAL HEALTH

Research increasingly identifies an important role for the gut microbiome in intergenerational health – studies find that the microbiome community shapes mothers' risk for disease during and after pregnancy, as well as her baby's health outcomes (1, 2). However, the impact of nutrition on the microbiome during pregnancy remains largely unexplored. **In this study, we find impacts of Chiyo's program on the gut microbiome that suggest improved digestion and lower risk for common pregnancy and postpartum complications.**

FINDINGS OVERVIEW

↘ **51%**
Decrease in abundance of *Ruminoccocus gnavus*

↘ **21%**
Decrease in abundance of *Collinsella*

↘ **91%**
Decrease in abundance of *Clostridioides difficile*

↗ **22%**
Overall feeling of symptom improvement



STUDY DESCRIPTION

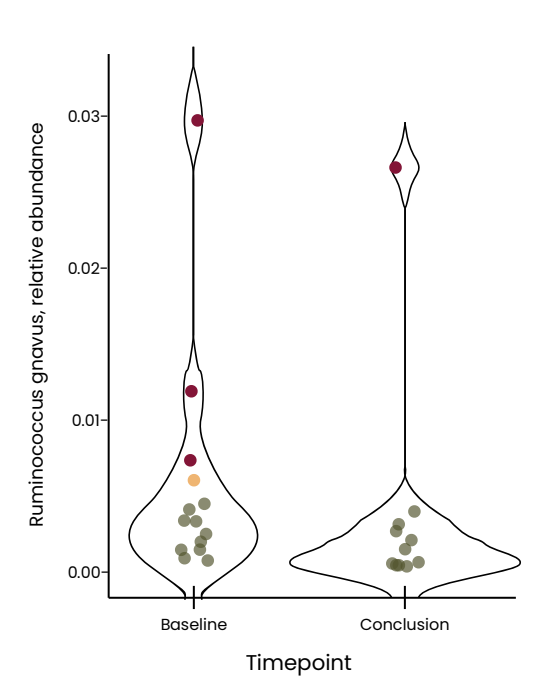
Chiyo partnered with Tiny Health to conduct a study testing the impact of Chiyo's Prenatal Nurture program on participants' gut microbiome and health symptoms. Participants received daily meals (breakfast, lunch, and dinner 7 days/week) from Chiyo's Prenatal Nurture Program for 6 weeks, completed surveys on their physical and mental health, stress, and sleep, at baseline (before starting the program), midpoint, and conclusion of the study, and collected poop samples for gut microbiome analysis at baseline and conclusion.

PARTICIPANTS' GUT MICROBIOMES SHOWED POSITIVE CHANGES AFTER CHIYO'S PRENATAL PROGRAM

After accounting for participants' diet quality at baseline and the number of days between their conclusion survey and microbiome sample, we found the abundance of several bacteria that have been linked to pregnancy complications, including *Ruminococcus gnavus, Collinsella*, and *Clostridioides difficile,* decreased after participants completed the Prenatal Nurture Program.

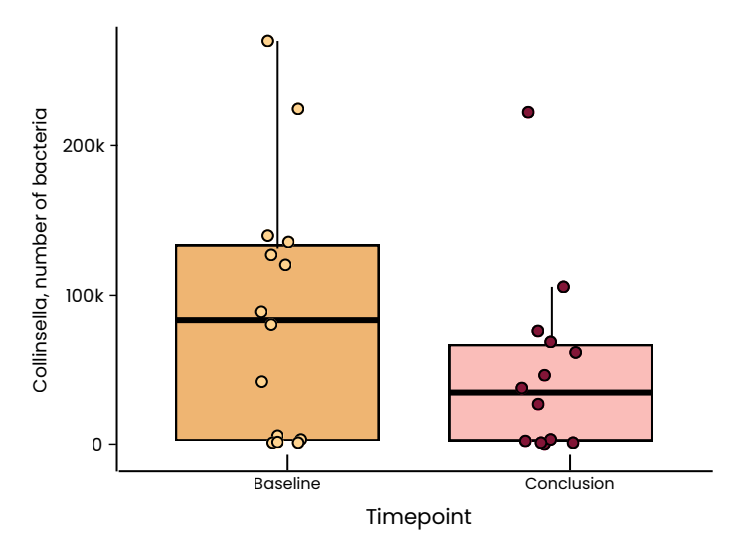
Ruminococcus gnavus is a bacterial species that has been linked to a higher risk of several pregnancy complications, including premature birth (3), pre-eclampsia (4), and gestational diabetes, where it is also correlated with higher fasting blood sugar in late pregnancy (5). Outside of pregnancy, higher abundance of *Ruminococcus gnavus* has also been associated with several diseases (e.g., migraine, depression, anxiety, IBS) in some studies, but not all (6).

As shown in the graph, there was a statistically significant decrease in *Ruminococcus gnavus* abundance after completing the program. This decrease was found in almost all participants (12 out of 13 analyzed) including most of those who had "borderline high" or "high" levels of *Ruminococcus gnavus* at baseline (orange and red dots). The **average decrease in abundance across the program was 51%.**



Graph Notes: Each dot represents a participant. Green = "typical" or "low" levels; Yellow = "borderline high" levels; Red = "high" levels. Color categories are based on typical population values for each individual by age.

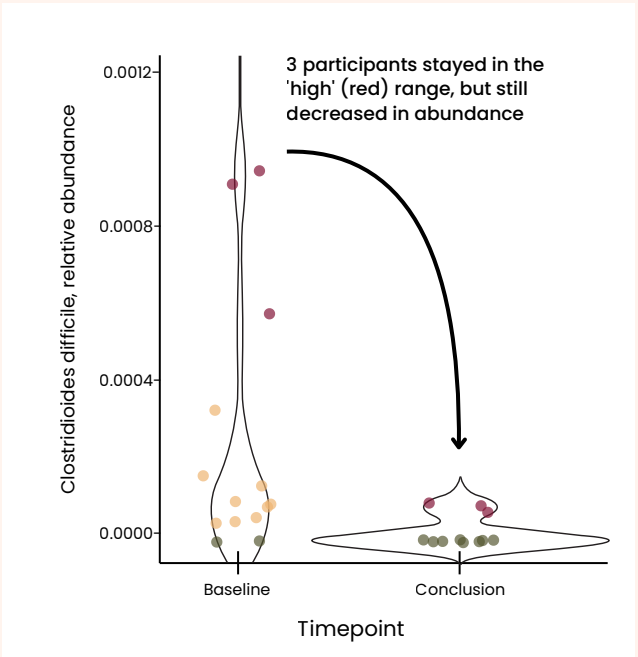
Collinsella is another bacterial genus that has been linked to blood sugar control in pregnancy, with higher abundance associated with more insulin resistance (7) and higher risk of gestational diabetes (8). In our study, there was a statistically significant decrease in *Collinsella* abundance from baseline to conclusion, with 9 out of the 13 participants (69%) showing a decrease. **The average decrease in abundance across the program was 21%.** Lower abundance of *Collinsella* was also significantly linked to better digestive health (more optimal bowel movement frequency and regularity).



Graph Notes: Each dot represents a participant. Thick black lines represent the median (middle) value at each timepoint, and each box captures the middle 50% of the data at that timepoint.

Note on the Pregnancy Context: Substantial gut microbiome changes happen across pregnancy (9). Generally, the microbiome shifts to a more inflammatory profile, with more dysbiosis, from the first to third trimester. So, the fact that we saw changes associated with lower pregnancy complications is another point to suggest that Chiyo's Prenatal Nurture Program had a positive impact.

Clostridioides difficile is a bacteria known for causing mild to severe gastrointestinal infections[1] (10). Though some people can have toxin-producing strains of this bacteria in their gut without any symptoms, antibiotic use can disturb the balance of the gut microbes in these individuals, enabling *Clostridioides difficile* to multiply and produce an infection (11). *Clostridioides difficile* infection during pregnancy is associated with higher risk of c-section delivery (12). As shown in the graph below, only 14% of participants did not have *Clostridioides difficile* in their gut microbiome (green dots) at the start of Chiyo's Prenatal Nurture Program, whereas **by the end, 85% did not have any *Clostridioides difficile* bacteria.**



Graph note: Each dot represents a participant. Green = no Clostridium difficile, "typical" or optimal levels; Yellow = "borderline high" levels; Red = "high" levels. Color categories are based on typical population values for each individual by age.

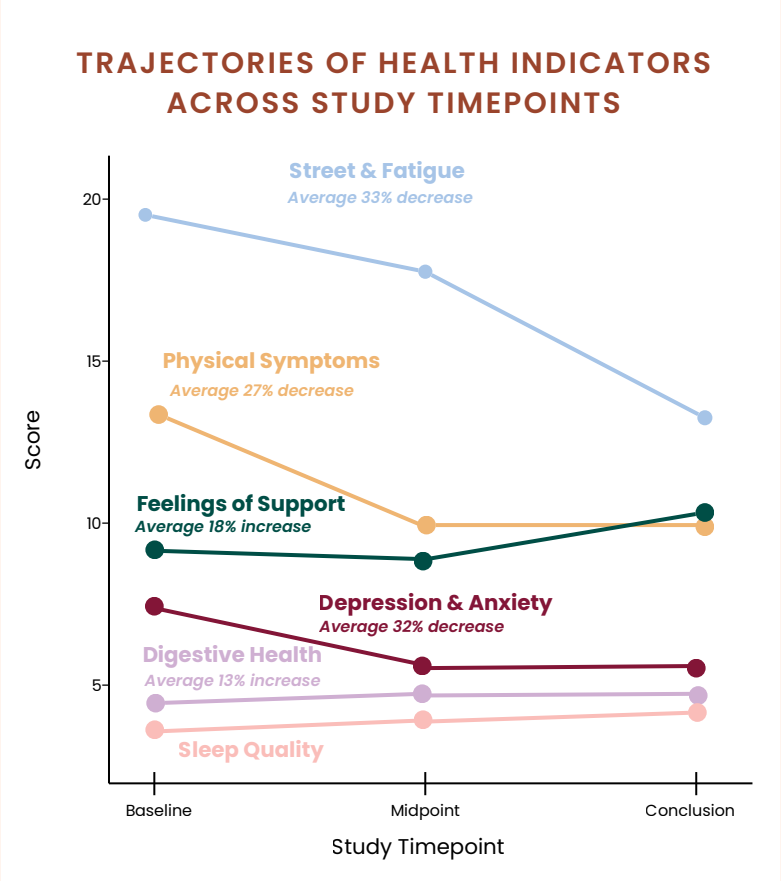
The average decrease in *Clostridioides difficile* across the program was 91%. These results suggest that the program reduced participants' vulnerability to infection and set them up for lower risk of delivery complications.

[1]There are several different strains of *Clostridioides difficile* bacteria: some release toxins and cause infections, while others don't (13). The microbiome testing method used in this study was not able to determine which type of strain was present. Future research should follow up on this finding to determine the strains involved.

PARTICIPANTS REPORTED BETTER PHYSICAL AND MENTAL HEALTH AFTER CHIYO'S PRENATAL PROGRAM

93% of respondents reported that they felt better – mentally, physically, and emotionally – after completing the Chiyo Prenatal Nurture Program. Of those who reported feeling better, the average reported overall improvement was 22%.

All health indicators tested, except for sleep quality, showed statistically significant improvements after participants started Chiyo's Prenatal Nurture Program. Average improvements from the start to the end of the program are shown in the graph to the right:



Graph Notes: Each dot represents the study average score at the given timepoint.

WHO PARTICIPATED IN THE STUDY?

Fifteen women in their second trimester of pregnancy (average of 20.78 weeks gestation at the start, range = 13.43 - 27.29 weeks) participated in the study. Thirteen of them provided data for gut microbiome analysis.

INSIGHTS FROM PARTICIPANT FEEDBACK:

Qualitative feedback highlights that participants appreciated the convenience and good taste of the program's meals, and felt better physically and mentally during the program.

> I will definitely be ordering from you guys postpartum! The meals were so good and diverse and I felt much better than usual sticking to the meal plan!
>
> — Holly* (30 weeks into pregnancy)

> When I stick to the plan all day, I feel much better, less bloated, and more clearheaded.
>
> — Lauren* (30 weeks into pregnancy)

**participant names were replaced with pseudonyms to preserve privacy.*



WHAT'S NEXT FOR THIS RESEARCH

As a research team, we are committed to conducting rigorous research on the impacts of our programs. In addition to what is shared in this white paper, we are planning to do additional analyses in partnership with Tiny Health comparing the microbiomes of study participants to a control group of pregnant women who did not complete Chiyo's Prenatal Nurture program, but who were at the same stage of pregnancy as study participants. Ultimately, we plan to publish our final results in a peer-reviewed journal so that they can be shared with the scientific community.

SPECIAL THANKS

We would like to express our deep appreciation for the women who participated in this study. Their dedication and willingness to share their insights – even during a demanding life stage – is a critical component of our work. We are profoundly inspired by their commitment to bettering not only their own health, but the health of other women and future generations.